STUDENT INNOVATION • IoT HEALTHCARE

SMART MEDICINE ASSISTANT

& Guardian Alert System

A cost-effective smart dispenser designed for safer, more reliable medication routines.

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DEMO-READY CONCEPT

1 THE PROBLEM

Why Medication Safety Matters

Medication non-adherence and dispensing errors create serious health risks, especially for senior citizens.

Medication Non-Adherence

Forgetfulness
Can lead to skipped doses.

Visual Decline
Makes identification and timing harder.

Complicated Schedules
Multi-dose routines increase confusion.

Risk of Double-Dosing
Accidental extra doses can be dangerous.

Limitations of Conventional Dispensers

- Often depend on high-speed home Wi-Fi.
- May lack physical tamper / compartment locks.
- Passive alarms may not reach remote caregivers.
- Missed doses can go unnoticed for too long.



Our Solution

One compact system connecting patient guidance, physical access control, sensing and caregiver escalation.

Guides the patient

Kannada • Hindi • English voice prompts

Controls access

Only the scheduled compartment unlocks

Verifies the dose

Opening + pill removal are checked

SMART MEDICINE ASSISTANT

SCHEDULED SLOT

LED + Voice
+ Servo Lock
+ Intake Sensors

Connects remotely

Hybrid GSM + Wi-Fi for SMS, calls and connected alerts

Escalates when missed

Guardian notification → automated call → audible alarm

What Makes It Innovative

Four layers of protection work together.

01

Regional Voice

Dosage instructions in Kannada, Hindi and English; guardians can record personal reminders.

02

Anti-Confusion Lock

Motorized servo physically seals non-scheduled compartments.

03

Dual Verification

Reed switch detects opening while IR/weight sensing checks medicine removal.

04

Hybrid Connectivity

GSM reduces dependence on mandatory home Wi-Fi and supports SMS / calls.

Core principle: reminder + physical control + verification + escalation

How It Works

A simple five-step journey from scheduled dose to verified intake.

1

SCHEDULE

DS3231 RTC keeps precise medication timing.

2

UNLOCK

Servo releases only the scheduled compartment.

3

GUIDE

LED + regional voice prompts the patient.

4

VERIFY

Reed + IR/weight sensing checks actual medicine removal.

5

ESCALATE

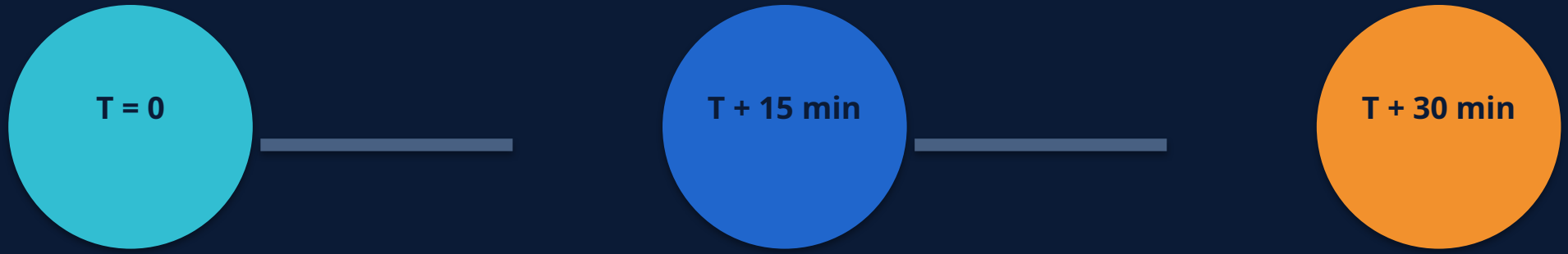
If missed, guardian alerts progress from push to auto-call.

SAFETY CHECK

The dose is considered complete only after compartment opening and medicine removal are detected.

Guardian Escalation

Three-stage response when the medicine is not removed.



PATIENT CUE

LED glow
Regional voice alert
Servo unlocks active slot

GUARDIAN PUSH

App notification / WhatsApp alert
to primary guardian

AUTO-CALL

IVR automated call + SMS
Guardian + backup contact
Loud unit alarm

Designed to reduce the chance that a missed dose stays unnoticed.

System Architecture

Hardware and functional layers from timing to alerts.

MAIN CONTROLLER

ESP32 / NodeMCU / Arduino + DS3231 RTC — Timing, sensor polling & communication

INTAKE VERIFICATION

IR optical pair / load cell + Reed switch — Door status + medicine extraction

AUDIO & DISPLAY

DFPlayer Mini + 3W speaker + 16×2 I2C LCD — Voice guidance, countdowns & timing

SAFETY LOCK

Micro Servo SG90 — Locks inactive compartments

TELEMETRY & ALERTS

SIM800L GSM + Wi-Fi transceiver — Telemetry, SMS & emergency calls

Reliability & Future Scope

The proposal extends beyond reminders to resilience and supply awareness.

01

Power resilience

18650 Li-ion rechargeable battery backup supports operation during grid failures.

02

Stock refill alerts

Monitors remaining dosage cycles and alerts the guardian 48 hours before the weekly supply runs out.

03

Future scope

Build further on the proposal's safety, connectivity and caregiver-support architecture.

NEXT DEVELOPMENT DIRECTION

- Improve enclosure design for easier cleaning and daily use
- Calibrate sensing for different pill sizes and weights
- Refine alert reliability and caregiver workflow
- Test the prototype across realistic medication schedules

Why This Matters

A practical bridge between patient independence and caregiver confidence.

AFFORDABILITY

Designed as an accessible healthcare IoT concept.

INDEPENDENCE

Regional voice guidance supports senior users.

SAFETY

GSM escalation keeps guardians informed remotely.

THE TAKEAWAY

We are not building just a pill box.
We are building a safety loop: schedule → guide → verify → escalate.

SMART MEDICINE ASSISTANT

Guardian Alert System

THANK YOU

Questions & Demonstration

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